

PRATICA DI MARE, Italy

WMO: 162450

Lat: 41.659N

Lon: 12.445E

Elev: 13

StdP: 101.17

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.8	(c) 2.1	(d) -8.0	(e) 1.9	(f) 5.9	(g) -5.1	(h) 2.5	(i) 5.4	(j) 12.0	(k) 11.0	(l) 10.8	(m) 11.1	(n) 4.0	(o) 30	(p) 0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.4	(c) 31.1	(d) 23.2	(e) 30.1	(f) 23.3	(g) 29.1	(h) 23.4	(i) 26.3	(j) 28.8	(k) 25.6	(l) 28.2	(m) 24.7	(n) 27.6	(o) 5.0	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 25.3	(b) 20.5	(c) 28.3	(d) 24.7	(e) 19.8	(f) 27.8	(g) 23.9	(h) 18.7	(i) 27.1	(j) 82.0	(k) 28.9	(l) 78.5	(m) 28.4	(n) 74.9	(o) 27.7	(p) 32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.8	7.6	-2.0	34.1	1.7	1.9	-3.3	35.5	-4.3	36.6	-5.3	37.6	-6.5	39.0
			-3.7	27.8	1.6	1.3	-4.9	28.8	-5.9	29.5	-6.8	30.3	-8.0	31.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	9.2	9.3	11.4	14.1	17.9	21.8	24.3	24.8	21.4	18.2	14.1	10.4		
	DBStd	6.03	2.75	2.74	2.33	2.18	2.21	2.41	1.71	1.81	2.22	2.37	3.10	2.93		
	HDD10.0	137	48	39	13	1	0	0	0	0	0	0	6	31		
	HDD18.3	1324	284	252	214	127	36	2	0	0	3	31	130	246		
	CDD10.0	2492	23	21	57	125	243	354	442	458	343	254	128	43		
	CDD18.3	637	0	0	0	1	21	107	184	199	96	27	2	0		
	CDH23.3	4693	0	0	0	5	92	748	1578	1746	479	44	1	0		
	CDH26.7	1140	0	0	0	1	7	144	416	518	54	1	0	0		
(14) Wind	WSAvg	3.4	3.7	3.7	3.6	3.4	3.1	3.0	3.0	2.9	3.1	3.2	3.8	3.8		
(15) Precipitation	PrecAvg	812	80	80	70	57	36	21	9	36	79	111	121	104		
	PrecMax	1112	143	216	163	179	130	65	53	131	255	246	361	326		
	PrecMin	522	3	13	6	5	2	3	0	0	1	12	12	29		
	PrecStd	161	42	55	44	42	32	17	13	31	69	79	83	65		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	17.2	20.1	23.2	27.3	31.1	32.1	33.9	29.8	25.9	22.1	18.2		
		MCWB	14.0	12.8	13.3	16.1	18.5	22.1	22.6	22.1	22.4	21.0	18.4	15.4		
	2%	DB	15.2	15.7	18.0	21.2	25.3	29.8	30.9	31.6	28.1	24.5	20.9	17.0		
		MCWB	12.9	12.4	13.3	16.1	19.0	22.2	22.8	23.2	22.3	20.5	17.5	14.6		
	5%	DB	14.5	14.8	16.8	20.0	24.1	28.2	29.9	30.2	27.0	23.3	19.9	16.0		
		MCWB	12.2	11.9	13.4	15.8	18.9	22.3	23.3	23.8	22.3	19.6	17.0	13.8		
	10%	DB	13.8	13.9	15.8	18.8	22.9	27.1	29.0	29.2	26.0	22.4	18.9	15.0		
		MCWB	11.6	11.2	13.0	15.4	18.4	21.9	23.6	23.9	21.6	19.0	16.3	12.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	14.5	15.6	18.7	22.1	25.6	27.1	27.6	25.6	22.7	19.5	16.1		
		MCDB	16.2	15.9	17.5	21.0	24.4	28.0	29.4	29.5	27.7	24.8	21.3	17.5		
	2%	WB	13.7	13.3	14.8	17.4	20.8	24.6	26.2	26.6	24.3	21.4	18.3	15.1		
		MCDB	15.1	14.8	16.6	20.0	23.3	27.4	28.7	29.0	26.5	23.4	20.1	16.6		
	5%	WB	12.8	12.6	14.1	16.5	19.9	23.7	25.5	25.8	23.4	20.5	17.4	14.2		
		MCDB	14.2	14.1	15.9	18.9	22.8	26.7	28.2	28.5	25.8	22.7	19.3	15.7		
	10%	WB	11.8	11.7	13.3	15.6	19.1	22.8	24.6	25.0	22.4	19.6	16.4	13.2		
		MCDB	13.4	13.4	15.1	17.9	22.0	25.9	27.5	28.0	25.0	21.9	18.4	14.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.7	7.8	8.1	8.6	8.7	8.4	8.4	8.0	7.7	7.0	7.0		
		MCDBR	7.1	7.8	8.6	9.7	10.2	9.9	9.6	9.8	8.5	7.9	7.0	6.7		
	5% WB	MCWBR	5.7	5.8	6.3	6.6	6.4	6.4	6.6	6.5	5.9	5.9	5.3	5.3		
		MCWBR	6.0	6.7	7.6	8.6	8.9	8.9	8.3	8.3	7.5	7.3	6.3	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.376	0.415	0.444	0.448	0.440	0.433	0.433	0.426	0.412	0.387	0.353		
		taud	2.420	2.329	2.215	2.159	2.196	2.272	2.295	2.306	2.319	2.321	2.319	2.410		
	Ebn at Noon		785	819	825	828	833	839	840	829	807	767	728	747		
		Edh at Noon	81	102	128	145	143	133	129	124	114	102	89	76		
(44) All-Sky Solar Radiation	RadAvg		1.76	2.71	3.98	5.33	6.52	7.39	7.52	6.55	4.79	3.27	2.01	1.60		
	RadStd		0.17	0.26	0.44	0.34	0.52	0.30	0.32	0.33	0.23	0.27	0.18	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-2.38	N/A	N/A	N/A	N/A	N/A	+128	+59
(47) Regional (16 neighbors)		+0.42	N/A	N/A	+0.53	N/A	N/A	N/A	-78	+135	+69

Nomenclature: See separate page